Partial Derivatives

$y = f(x)$ → single independent variable.

↓
Dependent variable

e.g. Area of a square = $4 \times (\text{sides})^2$

∴ Area of a square depends only on one variable quantity.

∴ $A = 4c^2$

∴ $y = f(c)$

Now, Area of a triangle = $\frac{1}{2} lb$

∴ Area of a triangle depends upon two variables quantities.

∴ $y = f(l, b)$

↓ ↓

2 independent variables.

For the partial derivatives, we introduce a new dependent variable u and two independent variables are x & y.

∴ $u = f(x, y)$ → Geometrically represents surface in three dimensional space

or $z = f(x, y)$

Functions of two or more variables:

The function $f(x, y)$ is called a real valued function of two or more variables if there are two or more independent variables.

e.g. Total surface area of a rectangular parallelopiped is $2(xy + yz + zx)$ which is function of three variables.

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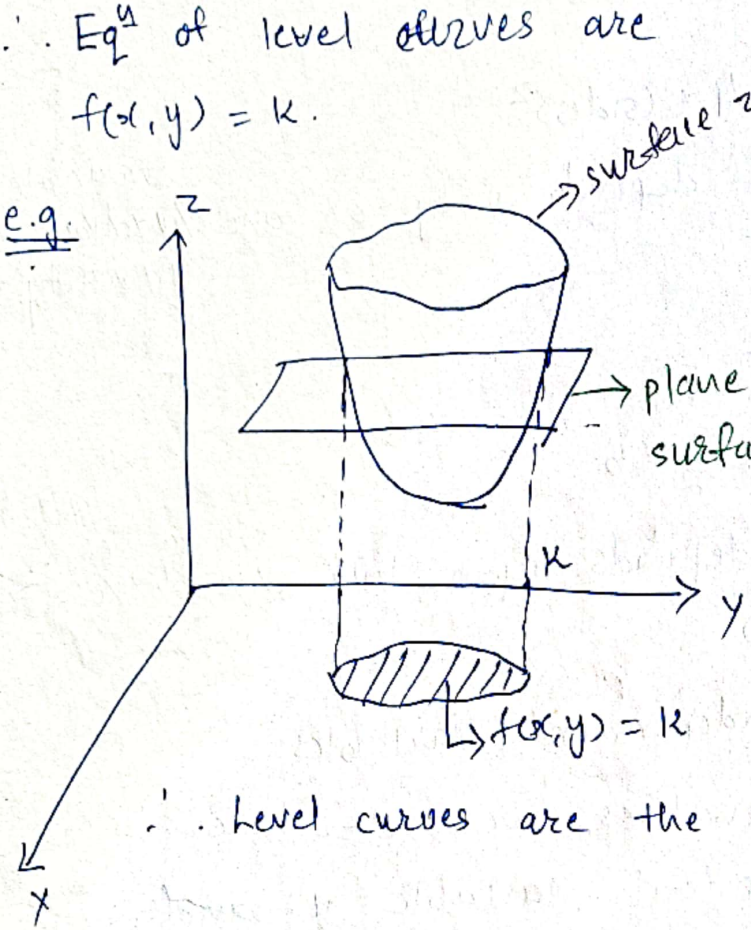
Graphs and level curves: xy-plane, xyz-surface

The level curves of the function $z = f(x, y)$ are two dimensional curves obtained by substituting $z = k$, where k is any number.

∴ Eqⁿ of level curves are

$f(x, y) = k$.

e.g.



∴ Level curves are the projection on xy-plane

e.g. Consider a function $f(x, y) = \sqrt{x^2+y^2}$

Let, $z = \sqrt{x^2+y^2}$.

Squaring both the sides

$z^2 = x^2+y^2$

→ The level curves for this function are found by substituting $z = k$.

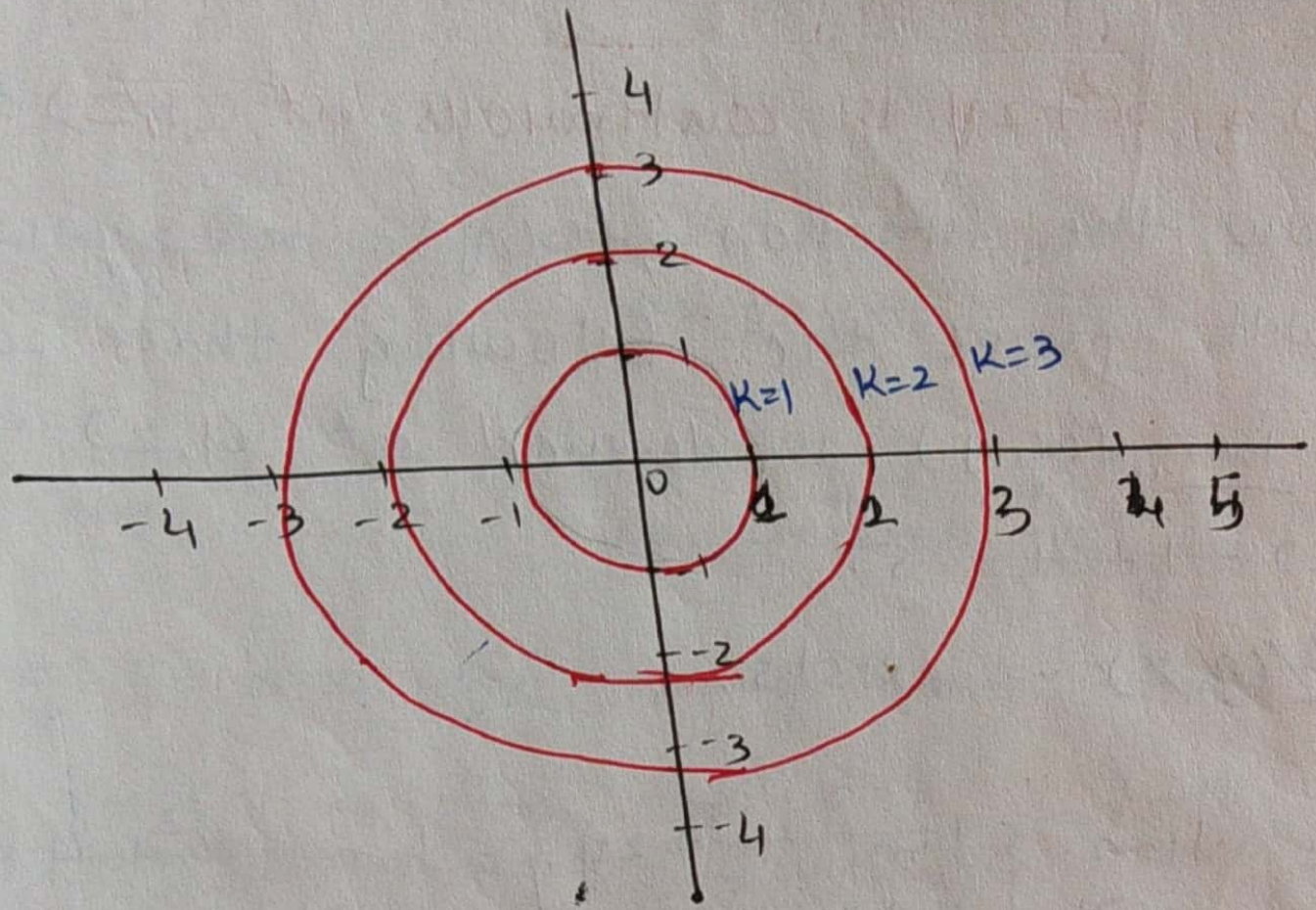
∴ $x^2+y^2 = k^2$, k is any number.

∴ Level curves are circles of radius k with centre at the origin.

→ We can graph level curves in two ways

1) graph them on surface
2) in a two dimensional axis system.

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The set of all points $(x, y, f(x,y))$ in space, for (x,y) in the domain of f, is called the graph of f.

- The graph of f is also called surface $z = f(x,y)$.

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Unit 5: Partial Derivatives

[2] Limit, Continuity of function of several variables

* Limit of a function of two variables

[let for $\epsilon > 0$ $\exists$ $\delta > 0$ ∋ $|f(x) - l| < \epsilon$;
whenever $|x - a| < \delta$

i.e. If the difference between $f(x)$ and fixed value l can be made as small as possible by taking the value of x sufficiently near to 'a'

→ Definition:

Let $f(x,y)$ be a function defined in the neighbourhood of a point (a,b). The function $f(x,y)$ is said to have limit 'L' as the point (x,y) approaches (a,b) if given $\epsilon > 0$ $\exists$ a number $\delta > 0$ ∋

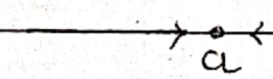
$|f(x,y) - l| < \epsilon$

whenever $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$.

→ This is denoted as $\lim_{(x,y)\to(a,b)} f(x,y) = L$

→ [In a function of one variable, if left limit and right limit of a function exist and are equal, then limit of a function exists.]

→ So, in a function of one variable 'x' approach to 'a' only in two directions either from the left or from right



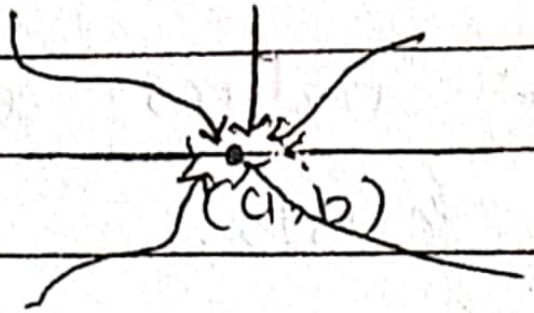
→ But a function of two variables, (x,y) approach to (a,b) in infinitely many ways

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and path of approaches may be straigh
line or curves.



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Limit :-

1) **δ-neighbourhood** -

The δ-neighbourhood of a point (a, b) in the xy-plane is a square with centre at (a, b) bdd. by four lines

$x = a - \delta$, $x = a + \delta$, $y = b - \delta$, $y = b + \delta$

i.e. $a - \delta \leq x \leq a + \delta$

$b - \delta \leq y \leq b + \delta$

2) **Neighbourhood of (a, b)** :-

Neighbourhood of a pt. (a, b) is also defined as circular disk with centre at (a, b) and of radius given by $(x-a)^2 + (y-b)^2 < \delta^2$

i.e. The distance between the points (x, y) & $(x_0, y$ in the plane is less if

$$\sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta.$$

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* Examples:

(1) Show that $\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$ (By def^n)

Sol^n: W.K.T.

for given $\epsilon > 0$ $\exists$ $\delta > 0$ ∋

$|f(x,y) - l| < \epsilon$; if $0 < \sqrt{(x-a)^2+(y-b)^2} < \delta$.

Here $f(x,y) = \frac{3x^2y}{x^2+y^2}$, $l = 0$, $a = 0, b = 0$

$\therefore \left|\frac{3x^2y}{x^2+y^2} - 0\right| < \epsilon$; $0 < \sqrt{x^2+y^2} < \delta$

$\Rightarrow \frac{3x^2|y|}{x^2+y^2} < \epsilon$; $0 < \sqrt{x^2+y^2} < \delta$

Now,

$\frac{x^2}{x^2} = 1 \Rightarrow \frac{x^2}{x^2+y^2} \leq 1$ $(y^2 \geq 0)$

$\Rightarrow \frac{3x^2|y|}{x^2+y^2} \leq 3|y| = 3\sqrt{y^2} < 3\sqrt{x^2+y^2}$

$\therefore \frac{3x^2|y|}{x^2+y^2} \leq 3\sqrt{x^2+y^2}$ But $\frac{3x^2|y|}{x^2+y^2} < \epsilon$

$\therefore \frac{3x^2|y|}{x^2+y^2} \leq 3\sqrt{x^2+y^2} < \epsilon$

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$\therefore\ 3\sqrt{x^2+y^2} < \epsilon$

$\Rightarrow \sqrt{x^2+y^2} < \epsilon/3$ but $\sqrt{x^2+y^2} < \delta$

i.e. If we choose $\delta = \epsilon/3$ then $\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$

(2) Show that $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$ (By defn

Sol^n: for given $\epsilon > 0$ $\exists$ $\delta > 0$ ∋

$|f(x,y) - l| < \epsilon$; $0 < \sqrt{(x-a)^2+(y-b)^2} < \delta$

$\therefore \left| \frac{xy}{\sqrt{x^2+y^2}} \right| < \epsilon$; $0 < \sqrt{x^2+y^2} < \delta$

Now,

$\frac{x^2}{x^2} = 1$

$\Rightarrow \frac{x^2}{x^2+y^2} \leq 1 \Rightarrow x^2 \leq x^2+y^2 \Rightarrow \sqrt{x^2} \leq \sqrt{x^2+y^2}$

$\Rightarrow \frac{|x|}{\sqrt{x^2+y^2}} \leq 1 \Rightarrow \frac{|x||y|}{\sqrt{x^2+y^2}} \leq |y| = \sqrt{y^2} \leq \sqrt{x^2+y^2}$

$\therefore \frac{|x||y|}{\sqrt{x^2+y^2}} \leq \sqrt{x^2+y^2}$ but $\frac{|x||y|}{\sqrt{x^2+y^2}} < \epsilon$

$\therefore \frac{|x||y|}{\sqrt{x^2+y^2}} \leq \sqrt{x^2+y^2} < \epsilon$

$\Rightarrow \sqrt{x^2+y^2} < \epsilon$, But $\sqrt{x^2+y^2} < \delta$

i.e. If we choose $\delta = \epsilon$ then $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$

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Working rule for evaluation of limit:

1) Evaluate limits

i) $\lim_{x \to a} \{ \lim_{y \to b} f(x,y) \}$ and

ii) $\lim_{y \to b} \{ \lim_{x \to a} f(x,y) \}$

If both limit values are equal then

$\lim_{(x,y) \to (a,b)} f(x,y)$ exists.

2) If $a=0$, $b=0$, evaluate limit along different path

Say $y = mx$ or $y = mx^n$ etc.

If all limit values are equal, then $\lim_{(x,y) \to (a,b)} f(x,y)$ exi

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* Evaluate the following limits :

(1) $\lim_{(x,y)\to(0,1)} \frac{x-xy+3}{x^2y+5xy-y^3} = \frac{0-(0)(1)+3}{0(1)+5(0)(1)-1^3} = -$

(2) $\lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$

$= \lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}} \times \frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}+\sqrt{y}}$

$= \lim_{(x,y)\to(0,0)} \frac{x(x-y)(\sqrt{x}+\sqrt{y})}{(x-y)}$

$= 0$

(3) $\lim_{(x,y)\to(0,\frac{\pi}{4})} \sec x \tan y = \sec 0 \tan\frac{\pi}{4} = 1$

(4) $\lim_{(x,y)\to(0,0)} \frac{x^4-y^2}{x^4+y^2}$

Sol^{n} :

Path 1 : x-axis i.e. $y=0$

$L_1 = \lim_{x\to0}\left\{\lim_{y\to0}\frac{x^4-y^2}{x^4+y^2}\right\} = \lim_{x\to0}\left\{\frac{x^4}{x^4}\right\} = \lim_{x\to}$

Path 2 : y-axis i.e. $x=0$

$L_2 = \lim_{y\to0}\left\{\lim_{x\to0}\frac{x^4-y^2}{x^4+y^2}\right\} = \lim_{y\to0}\left\{\frac{-y^2}{y^2}\right\} = \lim_{y\to}$

Here $L_1 \neq L_2$

$\therefore$ limit does not exist.

[we can say that f possesses different limit along different paths. $\therefore$ limit doesn't e

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(5) $\lim_{(x,y)\to(0,0)} \frac{xy}{y^2-x^2}$

Sol^{n}:

Path 1 : x-axis i.e. $y=0$

$$L_1 = \lim_{x\to 0}\left\{\lim_{y\to 0}\frac{xy}{y^2-x^2}\right\} = \lim_{x\to 0}\left\{\frac{x(0)}{0-x^2}\right\} = 0$$

Path 2 : y-axis i.e $x=0$

$$L_2 = \lim_{y\to 0}\left\{\lim_{x\to 0}\frac{xy}{y^2-x^2}\right\} = \lim_{y\to 0}\left\{\frac{(0)y}{y^2-0}\right\} = 0$$

Path 3 : $y = mx$

$$L_3 = \lim_{(x,y)\to(0,0)}\frac{xy}{y^2-x^2} = \lim_{x\to 0}\frac{x(mx)}{m^2x^2-x^2}$$

$$= \lim_{x\to 0}\frac{mx^2}{x^2(m^2-1)}$$

$$= \frac{m}{m^2-1}$$

In this path limit depends on m & m is not fixed.

$\therefore L_1 = L_2 \neq L_3$

$\therefore$ limit does not exist.

(6) $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$

Sol^{n}:

Path 1 : x-axis, i.e. $y=0$

$$L_1 = \lim_{x\to 0}\left\{\lim_{y\to 0}\frac{x^2y}{x^4+y^2}\right\} = \lim_{x\to 0}\left\{\frac{x^2(0)}{x^4+(0)^2}\right\} = 0$$

Path 2 : y-axis, i.e. $x=0$

$$L_2 = \lim_{y\to 0}\left\{\lim_{x\to 0}\frac{x^2y}{x^4+y^2}\right\} = \lim_{y\to 0}\left\{\frac{(0)^2y}{(0)^4+y^2}\right\} = 0$$

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Path 3 : $y = mx$ (line)

$$L_3 = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2y}{x^4+y^2} = \lim_{x\to 0} \frac{x^2(mx)}{x^4+m^2x^2}$$

$$= \lim_{x\to 0} \frac{mx^3}{x^2(x^2+m^2)}$$

$$= \lim_{x\to 0} \frac{mx}{x^2+m^2} = 0$$

Path 4 : $y = mx^2$ (curve)

$$L_4 = \lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2} = \lim_{x\to 0} \frac{x^2 \cdot mx^2}{x^4+m^2x^4}$$

$$= \lim_{x\to 0} \frac{x^4 \cdot m}{x^4(1+m^2)}$$

$$= \frac{m}{1+m^2}$$, which depends on m

$\therefore L_1 = L_2 = L_3 \neq L_4$

$\therefore$ limit does not exist.

(7) $\lim_{(x,y)\to(0,0)} \frac{x^2y^4}{(x^2+y^4)^2}$

Soln :

Path 1 : x-axis, i.e. $y = 0$

$$L_1 = \lim_{x\to 0} \left\{ \lim_{y\to 0} \frac{x^2y^4}{(x^2+y^4)^2} \right\} = \lim_{x\to 0} \left\{ \frac{x^2(0)}{(x^2+0)^2} \right\}$$

Path 2 : y-axis i.e. $x = 0$

$$L_2 = \lim_{y\to 0} \left\{ \lim_{x\to 0} \frac{x^2y^4}{(x^2+y^4)^2} \right\} = \lim_{y\to 0} \left\{ \frac{(0)y^4}{(0+y^4)^2} \right.$$

Path 3 : $y = mx$ (line)

$$L_3 = \lim_{(x,y)\to(0,0)} \frac{x^2y^4}{(x^2+y^4)^2} = \lim_{x\to 0} \frac{x^2 \cdot m^4x^4}{(x^2+m^4x^4)^2}$$

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$$= \lim_{x\to 0} \frac{m^4 x^6}{x^4(1+m^4x^2)^2} = \lim_{x\to 0} \frac{m^4 x^2}{(1+m^2x^2)^2} = 0$$

Path 4 : $y = mx^2$ (curve)

$$L_4 = \lim_{(x,y)\to(0,0)} \frac{x^2y^4}{(x^2+y^4)^2}$$

$$= \lim_{x\to 0} \frac{x^2 \cdot m^4x^8}{(x^2+m^4x^8)^2}$$

$$= \lim_{x\to 0} \frac{m^4 x^{10}}{x^4(1+m^4x^6)^2}$$

$$= \lim_{x\to 0} \frac{m^4 x^6}{(1+m^4x^6)^2} = 0$$

Path 5 : $x = my^2$ (curve)

$$L_5 = \lim_{(x,y)\to(0,0)} \frac{x^2y^4}{(x^2+y^4)^2}$$

$$= \lim_{y\to 0} \frac{m^2y^4 \cdot y^4}{(m^2y^4+y^4)^2}$$

$$= \lim_{y\to 0} \frac{m^2 y^8}{y^8(m^2+1)^2}$$

$$= \frac{m^2}{(m^2+1)^2}$$, which depends upon m

$\therefore L_1 = L_2 = L_3 = L_4 \neq L_5$

$\therefore$ limit does not exist.

(8) $\lim_{(x,y)\to(0,0)} \frac{x-y}{x^2+y^2}$

Sol^{n} :

Path 1 :

$$L_1 = \lim_{x\to 0} \left\{ \lim_{y\to 0} \frac{x-y}{x^2+y^2} \right\} = \lim_{x\to 0} \left\{ \frac{x}{x^2} \right\} = \lim_{x\to 0} \frac{1}{x} = \infty$$

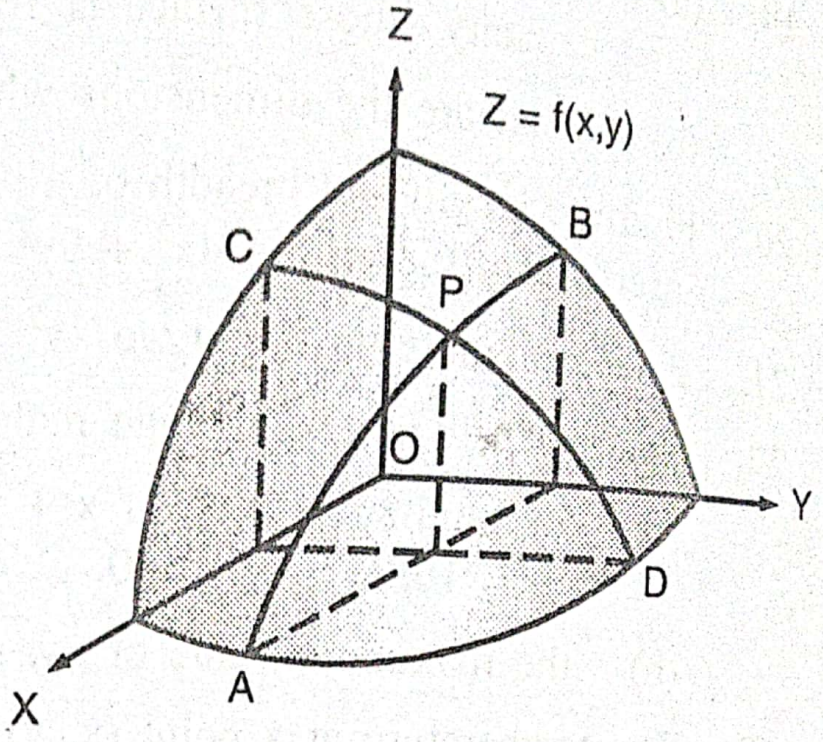
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6.7 Geometrical Interpretation of Partial Derivatives

Geometrically $z = f(x, y)$ represents a surface. Let a plane $y = y_0$ parallel to xz–plane pass through P cutting the surface along the curve APB. The equation of this curve is thus

$$z = f(x, y),\ y = y_0$$

All the points on this curve satisfy these two equations. As the point P moves along this curve its z–coordinate varies with x, while its y–coordinate remains constant.



Hence the rate of change of z with respect to x at P is equal to the **slope of the tangent to this curve at P measured relative to the axis of x.**

It is denoted by $\frac{\partial z}{\partial x}$ and has the same meaning as $\frac{dz}{dx}$.

Similarly $\frac{\partial z}{\partial y}$ at P is equal to the slope of the tangent to the curve CPD measured relative to the axis of y.

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